

Microprocessor

CSIT — 2nd Semester — 2082 — Answer Sheet
Subject Code: CSC167

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain instruction cycle, machine cycle and T-states. Draw timing diagram of instruction cycle of STA instruction with brief description.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Draw block diagram of 8086 microprocessor and explain its main two functional units. Differentiate between Real Address Mode and Protected Virtual address mode of 80286 microprocessor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the working of PUSH and POP instruction with suitable example. Write an assembly language program using 8085 microprocessor instructions to divide 34h by 08h, store the remainder in memory location 8050h and quotient in memory location 8051h.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Differentiate between Isolated mapped and Memory Mapped I/O. Explain basic DMA operation in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is flag? Explain all the flags available in 8085 microprocessor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write an assembly language program for 8086 microprocessor to convert the string "microprocessor is programmable" to uppercase.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is the use of AD7-AD0 in 8085 microprocessor? Explain address de-multiplexing process in 8085 microprocessor with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is interrupt? Explain 8259 interrupt controller in brief with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain Register Organization in 80386 microprocessor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Draw a logic diagram showing generation of memory and I/O read/write control signals in 8085 microprocessor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is Descriptor? How logical address is converted to physical address using Descriptor table with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a) Operating Modes of 8255 PPI b) Harvard Architecture

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.